Evaluating Knowledge Graph-Enhanced Context for Multiple-Choice Question Answering

1st Maryam Ghanbari
Computer Science Department
Brock University, Canada
mghanbari@brocku.ca

2nd Renata Dividino
Computer Science Department
Brock University, Canada
rdividino@brocku.ca

2025 IEEE International Conference on Knowledge Graph (ICKG) | 979-8-3315-6689-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICKG66886.2025.00020

***Abstract*—Despite their impressive capabilities, Large Language Models (LLMs) struggle to access knowledge not encoded during pre-training. In-context learning (ICL) addresses this limitation by embedding relevant information directly in the prompt, enabling LLMs to use external knowledge without updating their parameters. Recent research has explored integrating knowledge from knowledge graphs (KGs) into prompts, as KGs offer structured and factual representations of concepts and their relationships. A common strategy involves identifying key concepts in the task, grounding them in the KG, extracting subgraphs that connect question and answer concepts, and incorporating the corresponding statements into the prompts. However, a major challenge in ICL is selecting appropriate knowledge, that is, information that supports the model's reasoning and improves performance, while minimizing irrelevant or noisy content that can reduce accuracy. This study investigates how the representation of knowledge in prompts and the relevance and scope of task-grounded KG knowledge affect LLM performance on multiple-choice question answering (MCQA) tasks. We compare triple-based versus path-based representations, assess relevance filtering strategies, and evaluate different knowledge processing approaches. Our findings show that path-based representations outperform triple-based approaches but are more sensitive to noise. While KG-based knowledge can enhance LLM reasoning, imprecise selection of relevant knowledge can degrade performance below the zero-shot baseline (where no additional context is provided), highlighting the challenge of integrating KG knowledge without task-specific selection strategies. Expanding the scope of extracted subgraphs increases recall (retrieving more relevant information) but reduces precision (as more noise is also included). These findings underscore the critical importance of balancing informativeness with noise reduction in KG-enhanced LLM systems. Source code is publicly available at https://github.com/maryam-ghanbari/KGSweetSpot.**

***Index Terms*—In-Context Learning, Knowledge Graphs, Large Language Models, Multiple-Choice Question Answering, Knowledge Graph Information Selection, Commonsense Tasks**

I. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across diverse natural language tasks, yet they face inherent limitations in accessing and utilizing specialized knowledge not explicitly encoded during training. These models are fundamentally bounded by their parametric knowledge, information embedded within their weights during pre-training, which may be incomplete, outdated, or insufficient for domain-specific reasoning tasks [1]. In-context learning (ICL) has emerged as a promising approach to address these knowledge limitations by embedding relevant information directly within prompts, allowing LLMs to leverage external knowledge without requiring model parameter updates [2, 3, 4]. Despite the effectiveness of this approach, challenges remain in determining which information should (or not) be included in prompts to maximize performance.

Previous work has demonstrated that injecting external knowledge from knowledge graphs (KGs) can support the reasoning ability of LLMs in knowledge graph question answering (KGQA) tasks. Many prior studies have focused on developing methods to incorporate such knowledge [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]. KGs provide structured representations of entities and their relationships, offering factual precision that unstructured text often lacks. By incorporating KG-derived information, LLMs can reduce incorrect reasoning and hallucinations without the need for costly model updates.

Nevertheless, in some cases, simply injecting knowledge from KGs that are semantically related to the question answering task into prompts is insufficient to improve performance. This limitation is particularly evident in commonsense reasoning tasks, where questions often require information from multiple KG subgraphs. In such cases, correct responses rarely follow from a single subgraph; instead, they depend on the combined information provided by several moderately relevant subgraphs that collectively support a coherent answer. Furthermore, determining the optimal amount of information to include in prompts presents significant challenges. Excessive information can introduce noise that reduces performance, while insufficient information can leave the model uninformed. This involves not only assessing the relevance of the information but also taking into account its size, the serialization of the prompt, and other factors.

To examine this hypothesis, our study explores the impact of KG-derived contextual knowledge on multiple-choice question answering (MCQA) performance on commonsense tasks, with a specific focus on four key dimensions: knowledge structure, relevance, extraction scope, and processing. We systematically investigate how various design choices affect reasoning performance, including the comparison between isolated triple representations and connected path structures, individual versus batch processing of reasoning paths, and the impact of varying information relevance thresholds. Through comprehensive experiments, we analyze the trade-offs between knowledge completeness and contextual precision to identify

979-8-3315-6689-0/25/$31.00 ©2025 IEEE
DOI 10.1109/ICKG66886.2025.00020

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:02:57 UTC from IEEE Xplore. Restrictions apply.

optimal configurations for different reasoning scenarios. Our empirical analysis reveals distinct patterns in how design choices interact with noise sensitivity, providing practical guidelines for implementing KG-enhanced prompting strategies that maximize reasoning accuracy while maintaining computational efficiency.

II. Related Work

LLM-based QA methods fall into two broad paradigms: those that rely solely on the model's internal knowledge [16, 17, 18], and those that incorporate external sources, notably KGs, to compensate for parametric limitations. Knowledge Graph Question Answering (KGQA) enhances factual accuracy and reasoning by combining textual and structured knowledge from KGs. Existing approaches to integrating KGs with LLMs can be grouped into four main categories:

Knowledge Injection during Pre-training or Fine-tuning These methods encode KG information into the model weights [7, 8, 9]. Models like KGT5 [19] and RoG [10] fine-tune on KG-grounded QA data, aligning outputs with KG structures.

Embedding-based Methods These represent KG entities and relations in latent space, using architectures tailored for reasoning. Early efforts like KV-Mem [20] evolved into models like EmbedKGQA [11], QA-GNN [5], GreaseLM [21], and TransferNet [22]. Though effective, these methods often require rigid, task-specific designs with limited flexibility.

Semantic Parsing (SP) SP-based approaches translate questions into logical forms (e.g., SPARQL) that are executed over the KG [23, 24]. These enable precise symbolic reasoning but depend heavily on annotated data [25].

Retrieval-Augmented Methods These retrieve relevant KG subgraphs during inference and append them to the input prompt [26, 27, 28]. PoG [29] constructs multi-hop paths and prunes irrelevant facts; GCR [30] integrates KG structure via graph-constrained decoding. Some methods frame LLMs as agents that explore KGs iteratively [12, 13, 14, 15], though this introduces latency [31]. While retrieval-based methods are model-agnostic and computationally efficient, their success depends on retrieving semantically relevant and complete facts. Poor retrieval can inject noise, harming answer quality.

III. Problem Definition

We consider a multiple-choice question $q \in Q$, where Q is the set of all possible questions, and each question is associated with a set of answer choices $C = \{a_1, a_2, \ldots, a_n\}$, where $a_c \in C$ is the unique correct answer and all others $C \setminus \{a_c\}$ are incorrect. A knowledge graph (KG) is defined as $G = (V, E)$, where V represents the set of entities and E represents the set of relations. Each edge $(h, r, t) \in E$, connecting a head entity h to a tail entity t via relation r, constitutes a *knowledge statement*, representing a factual or semantic assertion. More generally, a knowledge statement can also correspond to a path $(h, r_1, v_1, r_2, \ldots, v_{n-1}, r_n, t)$ connecting two entities h and t through a sequence of intermediate nodes $v_1, \ldots, v_{n-1}$ and relations $r_1, \ldots, r_n$, thereby capturing multi-step relational information within the graph.

A knowledge extraction mechanism retrieves a set of subgraphs $\mathcal{G}_q = \{G_{q_1}, \ldots, G_{q_k}\} \subseteq G$, containing information about KG concepts grounded in and linked to the question q and answer candidates C, via:

$$\text{extract}(G, q, \{a_1, \ldots, a_n\}) : G \times Q \times C \to \mathcal{G}_q.$$

The subgraph extraction is performed for a pair question concept $q \in V$ and an answer choice concept $a \in V$. The subgraph includes the seed concepts and all intermediate concepts on any shortest path of length at most $k = 2$ connecting them. Formally:

$$G_{q,a} = \{v \in V \mid d(q, v) + d(v, a) \leq k\},$$

where $d(x, y)$ denotes the shortest-path distance between concepts x and y in the KG. The extracted subgraphs $\mathcal{G}_q$ are combined with q and C using an encoding function Γ to form an augmented prompt:

$$p = \Gamma(\mathcal{G}_q, q, \{a_1, ..., a_n\}),$$

which serves as input for an LLM. The model's performance is evaluated using a metric $\rho(p, a_c)$, i.e., as accuracy, measuring its ability to select the correct answer given the prompt.

This paper investigates how the following dimensions of KG-derived context affect performance in MCQA:

Knowledge Serialization Format: We compare two function Γ encoding: triple-based and path-based encodings. For instance, triple-based encoding with these two tripled `(security, RelatedTo, vault)` and `(vault, AtLocation, bank)` is represented in the path-based format as `(security → RelatedTo → vault → AtLocation → bank)`. We hypothesize that path-based representations improve the model's ability to select correct answers by preserving semantic continuity.

Knowledge Extraction Scope: We analyze the impact of extract$(G, q, \{a_1, \ldots, a_n\})$ expanding KG traversal from 1-hop to 2-hop neighbors, examining how extending the search radius affects the quality and quantity of retrieved knowledge. We hypothesize that while higher hops retrieval may increase coverage, it may also introduce noise and reduce precision.

Knowledge Relevance: Given the KG concepts grounded in the question q and answer candidates C, we define *relevant knowledge* as subgraphs centered on concepts from the question or the correct answer a_c, and *irrelevant knowledge* as subgraphs centered on concepts from the question and the incorrect answers $C \setminus \{a_c\}$. We compare prompts augmented with (i) relevant knowledge only, (ii) irrelevant knowledge, and (iii) mixed content. Our goal is to understand how the presence of accurate cues affects model behavior and whether irrelevant or excessive content misleads the model.

Knowledge Processing Strategy (Grouping): We study how varying the batch size of knowledge units (e.g., individual triples vs. grouped triple sets) fed into the LLM influences performance. Grouping may enrich context, but excessive aggregation could overwhelm the model.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:02:57 UTC from IEEE Xplore. Restrictions apply.

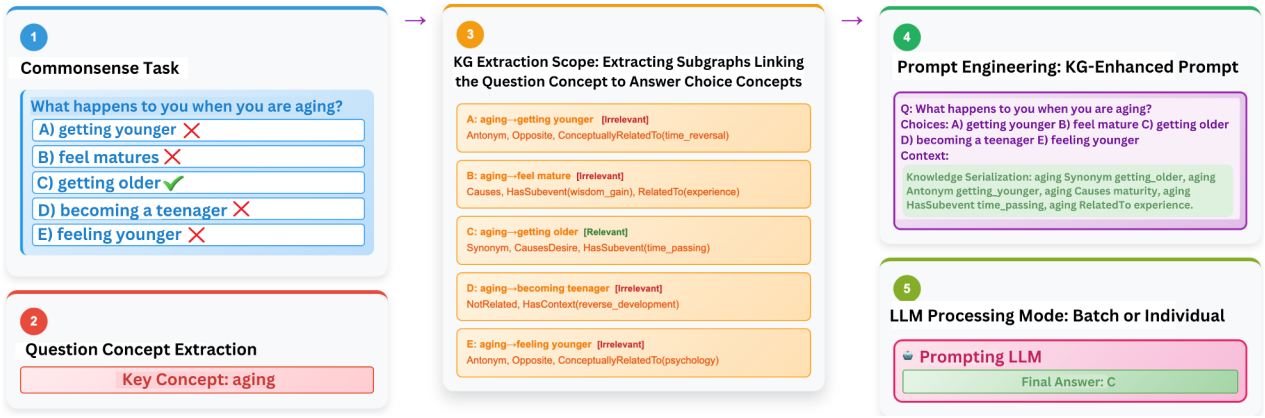


Fig. 1. Pipeline for evaluating KG-enhanced prompts. A sample MCQ is processed by identifying its key concepts, grounding each concept and each answer choice in the KG, and extracting subgraphs that connect them. The subgraphs are then converted into text according to a specific representation format and combined with the question and answer candidates to form an augmented prompt. The prompt is used by a LLM to generate the final answer. This study investigates how different strategies for incorporating KG-derived context influence LLM performance on MCQA tasks, focusing on four key dimensions: subgraph prompt representation, knowledge relevance, extraction scope, and LLM processing modes.

IV. Experiment Details

A. Dataset

MCQA Tasks: We investigate how the **structure**, **scope**, **processing**, and **relevance** of contextual knowledge from KG influence the performance on the CommonsenseQA (CSQA) dataset [32], a widely used benchmark for multiple-choice commonsense reasoning. The dataset contains 12,102 questions, each accompanied by five candidate answers, only one of which is correct. Answering these questions typically requires implicit commonsense knowledge that is not explicitly stated in the question or choices.

KG: We use ConceptNet [33], a general-domain commonsense KG comprising 799,273 nodes and 2,487,810 edges. Each node represents a concept, and edges capture commonsense relations such as `AtLocation`, `UsedFor`, and `CapableOf`.

B. Subgraph Knowledge Extraction

Following [5, 34], we adopt a multi-hop path extraction approach. Entities are identified from the question q and each answer choice $a_i \in C$ using simple string matching. Let $\{e_1^q, \ldots, e_m^q\}$ denote question entities, and $\{e_1^{a_1}, \ldots, e_l^{a_n}\}$ denote answer entities. We then extract all paths up to k-hops ($k = 1, 2$) in the KG that connect any question entity to any answer entity. For example, for the question *"A revolving door is convenient for two-direction travel, but it also serves as a security measure at a what?"*, with $k = 1$, and assuming the question entities are `revolving door` and `security`, and `bank` is an answer entity, retrieved statements include: `(revolving door → AtLocation → bank)` and `(security → RelatedTo → bank)`.

C. Subgraph Prompt Encoding

The way subgraphs $\mathcal{G}_q$ are encoded into the prompt $p = \Gamma(\mathcal{G}_q, q, C)$ significantly affects the model's ability to reason effectively [35]. Even with highly relevant knowledge, poor encoding can obscure connections between question and answers. We evaluate two encoding schemes: path-based representation (as employed in [5, 34]), and triple-based representation (as employed in [28]). We refer to each path or triple as a distinct knowledge statement.

D. Performance Metrics

Accuracy $\rho(p, a_c)$ is defined as the proportion of times the LLM correctly selects the answer a_c from the set of answer choices C, where a_c is the ground-truth correct answer. The performance metric is given by:

$$\rho(p, a_c) = \frac{\text{Number of correct selections}}{\text{Total number of questions}}$$

E. Inference Models

We leverage four inference models to process KG-augmented prompts: UnifiedQA-large, UnifiedQA-3b [36] Flan-T5-small, and Flan-T5-large [37].

We adopt and extend a knowledge integration strategy introduced in prior work [16], in which knowledge statements are incorporated into the input prompt to guide answer selection. Given a set of knowledge statements $M = \{k_1, k_2, \ldots, k_m\}$, an original question q, and answer choices $C = \{a_1, a_2, \ldots, a_n\}$, we construct augmented prompts by concatenating knowledge with the question and answer choices: $p_i = [k_i \,\|\, q \,\|\, C]$ for $i = 1, \ldots, m$, where $[\cdot \,\|\, \cdot]$ denotes textual concatenation using newline delimiters or other formatting tokens. Due to input length constraints, it is often infeasible to encode the entire set M within a single prompt. To address this, we explore two prompting strategies:

Single Knowledge Processing Each knowledge statement k_i is used to construct an individual prompt: $p_i = [k_i \,\|\, q \,\|\, C]$ for $i = 1, \ldots, m$. The model independently scores each answer candidate using each prompt, enabling it to focus on the impact of one knowledge statement at a time.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:02:57 UTC from IEEE Xplore. Restrictions apply.

TABLE I

ACCURACY WHEN PROVIDING 1-HOP KNOWLEDGE FILTERED TO INCLUDE ONLY STATEMENTS DIRECTLY LINKED TO QUERY CONCEPTS AND CORRECT ANSWER CHOICES (I.E., RELEVANT KNOWLEDGE), COMPARING INDIVIDUAL PROCESSING VERSUS BATCHES OF 2-10 STATEMENTS. PROVIDING RELEVANT KNOWLEDGE CONSISTENTLY IMPROVED MODEL ACCURACY ACROSS ALL TESTED MODELS. BATCH PROCESSING OF KNOWLEDGE STATEMENTS DID NOT PROVIDE PERFORMANCE BENEFITS COMPARED TO INDIVIDUAL STATEMENT PROCESSING.

Knowledge	**Models**			
	UnifiedQA-large	UnifiedQA-3b	Flan-T5-small	Flan-T5-large
No Knowledge	60.4%	71.9%	37.4%	84.6%
single	68%	83.6%	78.9%	96.6%
group of 2	$67.92\% \pm 0.22\%$	$84.46\% \pm 0.09\%$	$80.16\% \pm 0.20\%$	$96.5\% \pm 0.14\%$
group of 3	$67.60\% \pm 0.19\%$	$84.58\% \pm 0.16\%$	$80.56\% \pm 0.17\%$	$96.46\% \pm 0.13\%$
group of 6	$67.86\% \pm 0.26\%$	$84.56\% \pm 0.36\%$	$80.54\% \pm 0.17\%$	$96.54\% \pm 0.11\%$
group of 10	$67.94\% \pm 0.25\%$	$84.78\% \pm 0.16\%$	$80.28\% \pm 0.33\%$	$96.52\% \pm 0.08\%$

TABLE II

ACCURACY WHEN PROVIDING 1-HOP KNOWLEDGE FILTERED TO INCLUDE ONLY STATEMENTS DIRECTLY LINKED TO QUERY CONCEPTS AND INCORRECT ANSWER CHOICES (I.E., IRRELEVANT KNOWLEDGE), COMPARING INDIVIDUAL PROCESSING VERSUS BATCHES OF 2-10 STATEMENTS. ACROSS MOST TESTED MODELS, PROVIDING IRRELEVANT KNOWLEDGE EITHER SIGNIFICANTLY DEGRADED ACCURACY OR, AT BEST, RESULTED IN NO PERFORMANCE IMPROVEMENT. BATCH PROCESSING OF KNOWLEDGE STATEMENTS DID NOT PROVIDE PERFORMANCE BENEFITS COMPARED TO INDIVIDUAL STATEMENT PROCESSING.

Knowledge	**Models**			
	UnifiedQA-large	UnifiedQA-3b	Flan-T5-small	Flan-T5-large
No Knowledge	60.4%	71.9%	37.4%	84.6%
single	55.28%	67.81%	22.77%	60.93%
group of 2	$56.35\% \pm 0.5\%$	$67.03\% \pm 0.10\%$	$18.41\% \pm 0.36\%$	$53.79\% \pm 0.39\%$
group of 3	$58.79\% \pm 0.43\%$	$67.64\% \pm 0.48\%$	$16.14\% \pm 0.63\%$	$50.75\% \pm 0.43\%$
group of 6	$60.79\% \pm 0.73\%$	$67.65\% \pm 0.43\%$	$13.94\% \pm 0.20\%$	$47.78\% \pm 0.73\%$
group of 10	$61.62\% \pm 0.29\%$	$67.12\% \pm 0.44\%$	$13.50\% \pm 0.28\%$	$46.50\% \pm 0.48\%$

TABLE III

ACCURACY WHEN USING BOTH RELEVANT AND IRRELEVANT 1-HOP KNOWLEDGE, COMPARING INDIVIDUAL PROCESSING VERSUS BATCHES OF 2-10 STATEMENTS. EVEN IN THE PRESENCE OF RELEVANT KNOWLEDGE, THE ADDITION OF IRRELEVANT KNOWLEDGE AS CONTEXT MAINTAINS PERFORMANCE AT OR DEGRADES IT BELOW THE ZERO-SHOT BASELINE (NO ADDITIONAL KNOWLEDGE PROVIDED).

Knowledge	**Models**			
	UnifiedQA-large	UnifiedQA-3b	Flan-T5-small	Flan-T5-large
No Knowledge	60.4%	71.9%	37.4%	84.6%
single	59.62%	73.96%	38.33%	75.92%
group of 2	$59.46\% \pm 0.43\%$	$74.35\% \pm 0.41\%$	$38.17\% \pm 0.42\%$	$74.79\% \pm 0.72\%$
group of 3	$58.64\% \pm 0.71\%$	$73.51\% \pm 0.49\%$	$38.28\% \pm 0.45\%$	$74.24\% \pm 0.41\%$
group of 6	$55.69\% \pm 0.70\%$	$72.64\% \pm 0.52\%$	$38.31\% \pm 1.07\%$	$73.35\% \pm 0.96\%$
group of 10	$53.38\% \pm 0.30\%$	$71.78\% \pm 0.45\%$	$37.53\% \pm 0.90\%$	$72.74\% \pm 0.62\%$

Chunked Knowledge Processing The knowledge set M is partitioned into K disjoint subsets $S_1, \ldots, S_K \subset M$, where $S_i \cap S_j = \emptyset$ for $i \neq j$. Each chunk forms a single prompt: $p_k = [\, S_k \,\|\, q \,\|\, C \,] \quad \text{for } k = 1, \ldots, K$. This allows the model to evaluate answer candidates with multi-statement context, while staying within token limits.

For each prompt p_t and candidate answer $a \in C$, the model computes a score $s(a \mid p_t)$, and these are aggregated across all prompts: $s(a) = \frac{1}{T} \sum_{t=1}^{T} s(a \mid p_t)$, where $T = m$ in the single knowledge setting or $T = K$ in the chunked setting. Unlike [16], which selects the answer based on the maximum score from the best single prompt, we take the mean across all prompts to incorporate signals from the full knowledge set. This strategy enables better analysis of how all knowledge contributes to the model's decision. The aggregated scores are normalized using softmax.

The models operate in a teacher-forcing mode during inference, meaning that for each candidate answer, we compute the conditional log-likelihood of the target sequence by providing the ground-truth tokens as decoder inputs. In this deterministic setup, there is no autoregressive decoding, and the loss is solely determined by the discrepancy between the model's predicted logits and the given label tokens. As a result, inference is reproducible and not sensitive to random seeds.

V. RESULTS

Effect of Knowledge Relevance Relevant knowledge consistently enhanced model accuracy across all tested models. Providing relevant 1-hop triples individually significantly improved performance compared to baseline, with UnifiedQA-large increasing from 60.4% to 68%, UnifiedQA-3b from 71.9% to 83.6%, Flan-T5-small from 37.4% to 78.9%, and Flan-T5-large from 84.6% to 96.6% (see Table I). Results are of similar nature in Table IV, and Table VII. Conversely, irrelevant knowledge reduced accuracy, demonstrating the

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:02:57 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

ACCURACY WHEN PROVIDING RELEVANT 1- AND 2-HOP KNOWLEDGE. PROVIDING RELEVANT KNOWLEDGE CONSISTENTLY IMPROVED MODEL ACCURACY ACROSS ALL TESTED MODELS, THOUGH WITH VARYING MAGNITUDES OF IMPROVEMENT. THIS CONFIGURATION SHOWED SMALLER PERFORMANCE GAINS THAN 1-HOP KNOWLEDGE (SEE TABLE I), INDICATING THAT 2-HOP EXTRACTION INTRODUCES DETRIMENTAL NOISE THAT REDUCES MODEL PERFORMANCE.

Knowledge	**Models**			
	UnifiedQA-large	UnifiedQA-3b	Flan-T5-small	Flan-T5-large
No Knowledge	60.4%	71.9%	37.4%	84.6%
single	60.77%	77.6%	59.62%	92.8%
group of 2	$60.36\% \pm 0.21\%$	$78.08\% \pm 0.34\%$	$69.09\% \pm 0.24\%$	$93.42\% \pm 0.23\%$
group of 3	$60.33\% \pm 0.17\%$	$78.00\% \pm 0.41\%\%$	$72.09\% \pm 0.48\%$	$93.22\% \pm 0.27\%$
group of 6	$60.26\% \pm 0.46\%$	$77.89\% \pm 0.50\%\%$	$73.91\% \pm 0.35\%$	$92.68\% \pm 0.23\%$
group of 10	$61.46\% \pm 0.31\%$	$78.95\% \pm 0.30\%\%$	$72.76\% \pm 0.27\%$	$91.89\% \pm 0.26\%$

TABLE V

ACCURACY WHEN USING ONLY IRRELEVANT 1- AND 2-HOP KNOWLEDGE. PROVIDING IRRELEVANT KNOWLEDGE AS CONTEXT DEGRADES SYSTEM PERFORMANCE BELOW THE ZERO-SHOT BASELINE (NO ADDITIONAL KNOWLEDGE PROVIDED). ALTHOUGH THIS CONFIGURATION INCREASES THE TOTAL AMOUNT OF RETRIEVED DATA, THE PERFORMANCE DEGRADATION REMAINS SIMILAR TO THE 1-HOP CONFIGURATION (SEE TABLE II). BATCH PROCESSING OF KNOWLEDGE STATEMENTS DID NOT PROVIDE PERFORMANCE BENEFITS COMPARED TO INDIVIDUAL STATEMENT PROCESSING.

Knowledge	**Models**			
	UnifiedQA-large	UnifiedQA-3b	Flan-T5-small	Flan-T5-large
No Knowledge	60.4%	71.9%	37.4%	84.6%
single	57.33%	71.91%	29.89%	75.92%
group of 2	$57.82\% \pm 0.14\%$	$71.67\% \pm 0.14\%$	$24.87\% \pm 0.22\%$	$72.19\% \pm 0.21\%$
group of 3	$59.05\% \pm 0.14\%$	$72.07\% \pm 0.33\%$	$22.74\% \pm 0.13\%$	$71.53\% \pm 0.19\%$
group of 6	$60.99\% \pm 0.41\%$	$72.15\% \pm 0.25\%$	$19.76\% \pm 0.27\%$	$71.02\% \pm 0.45\%$
group of 10	$61.70\% \pm 0.23\%$	$72.05\% \pm 0.60\%$	$18.30\% \pm 0.41\%$	$70.11\% \pm 0.72\%$

TABLE VI

ACCURACY WHEN USING BOTH RELEVANT AND IRRELEVANT 1- AND 2-HOP KNOWLEDGE. SIMILAR TO THE 1-HOP CONFIGURATION (SEE TABLE III), EVEN IN THE PRESENCE OF RELEVANT KNOWLEDGE, THE ADDITION OF IRRELEVANT KNOWLEDGE AS CONTEXT MAINTAINS PERFORMANCE AT OR DEGRADES IT BELOW THE ZERO-SHOT BASELINE.

Knowledge	**Models**			
	UnifiedQA-large	UnifiedQA-3b	Flan-T5-small	Flan-T5-large
No Knowledge	60.4%	71.9%	37.4%	84.6%
single	58.23%	73.05%	36.45%	80.84%
group of 2	$58.53\% \pm 0.17\%$	$73.03\% \pm 0.21\%$	$36.06\% \pm 0.13\%$	$80.10\% \pm 0.28\%$
group of 3	$58.91\% \pm 0.54\%$	$72.86\% \pm 0.12\%$	$35.27\% \pm 0.19\%$	$80.29\% \pm 0.25\%$
group of 6	$59.54\% \pm 0.17\%$	$73.52\% \pm 0.05\%$	$35.27\% \pm 0.48\%$	$80.67\% \pm 0.25\%$
group of 10	$58.59\% \pm 0.29\%$	$74.37\% \pm 0.29\%$	$34.32\% \pm 0.25\%$	$80.51\% \pm 0.14\%$

detrimental impact of noisy inputs (see Tables II, V, and VIII). When relevant and irrelevant knowledge were combined (see Tables III, VI, and IX), performance improvements ranged from negligible to marginal, indicating that irrelevant information substantially undermines the benefits of relevant knowledge. This confirms that relevant knowledge boosts performance while irrelevant knowledge causes sharp decline, with incorrect statements misleading the model even when mixed with correct information.

Impact of Knowledge Structure Path-based knowledge shown greater accuracy improvements than triple-based, especially for relevant information. With relevant 1+2-hop knowledge, UnifiedQA-large, UnifiedQA-3b, Flan-T5-small and T5-large achieved 67.98%, 82.88%, 73.46%, and 96.56% accuracy using paths (Table VII) versus 60.77%, 77.6%, 59.62%, and 92.8% with triples (Table IV). However, irrelevant path-formatted knowledge caused sharper accuracy drops than irrelevant triples, suggesting that richer context can amplify negative effects of irrelevant information (Table VIII). This confirms that paths provide richer semantic structure and more effectively capture question-answer relationships, but also amplify the disruptive effects of irrelevant information.

Effect of Knowledge Extraction Scope Increasing extraction from 1-hop to 1+2-hop yielded mixed results. While some models showed moderate enhancement, others experienced performance decline. For example, Flan-T5-small accuracy decreased from 78.9% (1-hop) to 73.46% (1+2-hop), showing that although 2-hop retrieval includes more comprehensive context, it introduces excessive irrelevant information that degrades model reasoning.

Knowledge Processing Strategies Grouping multiple statements initially improved performance but plateaued after merging more than two or three statements. Contrary to expectations, this results suggest that LLMs are relatively insensitive to grouping strategies and that grouping alone does not enhance contextual understanding.

VI. CONCLUSION

This study examined how different strategies for integrating KG-derived context affect the performance of LLMs on

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:02:57 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

ACCURACY WHEN PROVIDING RELEVANT 1-HOP AND 2-HOP KNOWLEDGE USING PATH-BASED REPRESENTATION. PROVIDING RELEVANT KNOWLEDGE CONSISTENTLY IMPROVED MODEL ACCURACY ACROSS ALL TESTED MODELS. PATH-BASED REPRESENTATION DEMONSTRATED SUPERIOR PERFORMANCE COMPARED TO TRIPLE-BASED REPRESENTATION IN BOTH 1-HOP AND COMBINED 1- AND 2-HOP CONFIGURATIONS (SEE TABLE I AND TABLE IV).

Knowledge	**Models**			
	UnifiedQA-large	UnifiedQA-3b	Flan-T5-small	Flan-T5-large
No Knowledge	60.4%	71.9%	37.4%	84.6%
single	67.98%	82.88%	73.46%	96.56%
group of 2	$66.67\% \pm 0.25\%$	$81.96\% \pm 0.21\%$	$71.79\% \pm 0.22\%$	$95.48\% \pm 0.19\%$
group of 3	$65.88\% \pm 0.33\%$	$81.55\% \pm 0.25\%$	$70.78\% \pm 0.43\%$	$94.38\% \pm 0.26\%$
group of 6	$67.27\% \pm 0.29\%$	$81.55\% \pm 0.21\%$	$67.81\% \pm 0.36\%$	$93.48\% \pm 0.28\%$
group of 10	$68.40\% \pm 0.30\%$	$81.82\% \pm 0.22\%$	$65.98\% \pm 0.55\%$	$92.56\% \pm 0.31\%$

TABLE VIII

ACCURACY WHEN USING ONLY IRRELEVANT 1- AND 2-HOP KNOWLEDGE USING PATH-BASED REPRESENTATION. PROVIDING IRRELEVANT KNOWLEDGE AS CONTEXT DEGRADES SYSTEM PERFORMANCE BELOW THE ZERO-SHOT BASELINE (NO ADDITIONAL KNOWLEDGE PROVIDED). ALTHOUGH THE SAME INFORMATION IS PROVIDED TO THE MODEL AS IN THE TRIPLE-BASED CONFIGURATION (SEE TABLE V), THE PERFORMANCE DEGRADATION IS MORE PRONOUNCED IN THE PATH-BASED CONFIGURATION.

Knowledge	**Models**			
	UnifiedQA-large	UnifiedQA-3b	Flan-T5-small	Flan-T5-large
No Knowledge	60.4%	71.9%	37.4%	84.6%
single	57.66%	69.62%	24.73%	62.49%
group of 2	$58.48\% \pm 0.14\%$	$69.40\% \pm 0.19\%$	$21.77\% \pm 0.07\%$	$65.01\% \pm 0.13\%$
group of 3	$59.11\% \pm 0.25\%$	$69.40\% \pm 0.37\%$	$20.95\% \pm 0.19\%$	$66.80\% \pm 0.11\%$
group of 6	$58.86\% \pm 0.29\%$	$68.77\% \pm 0.42\%$	$20.46\% \pm 0.19\%$	$67.06\% \pm 0.48\%$
group of 10	$58.12\% \pm 0.53\%$	$68.39\% \pm 0.41\%$	$20.03\% \pm 0.25\%$	$66.21\% \pm 0.49\%$

TABLE IX

ACCURACY WHEN USING BOTH RELEVANT AND IRRELEVANT 1- AND 2-HOP HOP KNOWLEDGE, USING PATH-BASED REPRESENTATION. SIMILAR TO THE TRIPLE-BASED CONFIGURATIONS (SEE TABLE III, AND TABLE VI), EVEN IN THE PRESENCE OF RELEVANT KNOWLEDGE, THE ADDITION OF IRRELEVANT KNOWLEDGE AS CONTEXT MAINTAINS PERFORMANCE AT OR DEGRADES IT BELOW THE ZERO-SHOT BASELINE.

Knowledge	**Models**			
	UnifiedQA-large	UnifiedQA-3b	Flan-T5-small	Flan-T5-large
No Knowledge	60.4%	71.9%	37.4%	84.6%
single	60.11%	72.8%	36.28%	76.9%
group of 2	$60.50\% \pm 0.17\%$	$72.9\% \pm 0.18\%$	$35.73\% \pm 0.07\%$	$78.44\% \pm 0.38\%$
group of 3	$60.44\% \pm 0.29\%$	$72.7\% \pm 0.20\%$	$34.86\% \pm 0.29\%$	$79.74\% \pm 0.18\%$
group of 6	$60.10\% \pm 0.35\%$	$73.0\% \pm 0.22\%$	$34.22\% \pm 0.37\%$	$79.74\% \pm 0.30\%$
group of 10	$59.20\% \pm 0.40\%$	$73.4\% \pm 0.30\%$	$34.07\% \pm 0.34\%$	$79.41\% \pm 0.30\%$

MCQA tasks. Our results demonstrate that relevant knowledge consistently improves accuracy across all tested models, with path-based representations significantly outperforming triple-based approaches in both 1-hop and combined 1- and 2-hop configurations. Including irrelevant knowledge consistently degrades system performance below the zero-shot baseline, regardless of whether it is presented alone or mixed with relevant information, emphasizing the critical challenge of implementing effective task-based selection strategies. Notably, path-based representations demonstrate lower resilience to noise compared to triple-based approaches, though triple-based approach performance degradation remains significant. This suggests that richer context can amplify negative effects of irrelevant information. Batch processing of knowledge statements provides no performance benefits over individual statement processing across all experimental configurations. These findings underscore the critical importance of balancing informativeness with noise reduction in KG-enhanced LLM systems. Since information relevance cannot be determined a priori, the primary challenge lies in developing context-aware selection mechanisms that accurately approximate relevance, given the detrimental impact of irrelevant knowledge on model performance. Without effective relevance filtering, the noise from irrelevant information can significantly degrade performance, making it preferable to forgo KG-augmented prompting entirely.

REFERENCES

[1] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," *Advances in neural information processing systems*, vol. 33, pp. 1877–1901, 2020.

[2] R. Ren and Y. Liu, "Towards understanding how transformers learn in-context through a representation learning lens," *Advances in Neural Information Processing Systems*, vol. 37, pp. 892–933, 2025.

[3] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, "Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing," *ACM computing surveys*, vol. 55, no. 9, pp. 1–35, 2023.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:02:57 UTC from IEEE Xplore. Restrictions apply.

[4] Q. Dong, L. Li, D. Dai, C. Zheng, J. Ma, R. Li, H. Xia, J. Xu, Z. Wu, T. Liu *et al.*, "A survey on in-context learning," *arXiv preprint arXiv:2301.00234*, 2022.

[5] M. Yasunaga, H. Ren, A. Bosselut, P. Liang, and J. Leskovec, "Qa-gnn: Reasoning with language models and knowledge graphs for question answering," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2021, pp. 535–546.

[6] L. Luo, Z. Zhao, C. Gong, G. Haffari, and S. Pan, "Graph-constrained reasoning: Faithful reasoning on knowledge graphs with large language models," *arXiv preprint arXiv:2410.13080*, 2024.

[7] T. Shen, Y. Mao, P. He, G. Long, A. Trischler, and W. Chen, "Exploiting structured knowledge in text via graph-guided representation learning," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, B. Webber, T. Cohn, Y. He, and Y. Liu, Eds. Online: Association for Computational Linguistics, Nov. 2020, pp. 8980–8994. [Online]. Available: https://aclanthology.org/2020.emnlp-main.722

[8] R. Wang, D. Tang, N. Duan, Z. Wei, X. Huang, J. ji, G. Cao, D. Jiang, and M. Zhou, "K-adapter: Infusing knowledge into pre-trained models with adapters," 2020. [Online]. Available: https://arxiv.org/abs/2002.01808

[9] Z. Zhang, X. Han, Z. Liu, X. Jiang, M. Sun, and Q. Liu, "Ernie: Enhanced language representation with informative entities," 2019. [Online]. Available: https://arxiv.org/abs/1905.07129

[10] L. LUO, Y.-F. Li, R. Haf, and S. Pan, "Reasoning on graphs: Faithful and interpretable large language model reasoning," in *The Twelfth International Conference on Learning Representations*, 2024.

[11] A. Saxena, A. Tripathi, and P. Talukdar, "Improving multi-hop question answering over knowledge graphs using knowledge base embeddings," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds. Online: Association for Computational Linguistics, Jul. 2020, pp. 4498–4507. [Online]. Available: https://aclanthology.org/2020.acl-main.412/

[12] E. Markowitz, A. Ramakrishna, J. Dhamala, N. Mehrabi, C. Peris, R. Gupta, K.-W. Chang, and A. Galstyan, "Tree-of-traversals: A zero-shot reasoning algorithm for augmenting black-box language models with knowledge graphs," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, L.-W. Ku, A. Martins, and V. Srikumar, Eds. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 12 302–12 319. [Online]. Available: https://aclanthology.org/2024.acl-long.665/

[13] J. Sun, C. Xu, L. Tang, S. Wang, C. Lin, Y. Gong, L. M. Ni, H.-Y. Shum, and J. Guo, "Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph," 2024. [Online]. Available: https://arxiv.org/abs/2307.07697

[14] Q. Zhao, H. Yang, Q. Song, X. Yao, and X. Li, "Knowpath: Knowledge-enhanced reasoning via llm-generated inference paths over knowledge graphs," 2025. [Online]. Available: https://arxiv.org/abs/2502.12029

[15] J. Jiang, K. Zhou, W. X. Zhao, Y. Song, C. Zhu, H. Zhu, and J.-R. Wen, "Kg-agent: An efficient autonomous agent framework for complex reasoning over knowledge graph," 2024. [Online]. Available: https://arxiv.org/abs/2402.11163

[16] J. Liu, A. Liu, X. Lu, S. Welleck, P. West, R. Le Bras, Y. Choi, and H. Hajishirzi, "Generated knowledge prompting for commonsense reasoning," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, S. Muresan, P. Nakov, and A. Villavicencio, Eds. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 3154–3169. [Online]. Available: https://aclanthology.org/2022.acl-long.225

[17] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," *ArXiv*, vol. abs/2205.11916, 2022. [Online]. Available: https://api.semanticscholar.org/CorpusID:249017743

[18] L. Wang, W. Xu, Y. Lan, Z. Hu, Y. Lan, R. K.-W. Lee, and E.-P. Lim, "Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds. Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 2609–2634. [Online]. Available: https://aclanthology.org/2023.acl-long.147/

[19] A. Saxena, A. Kochsiek, and R. Gemulla, "Sequence-to-sequence knowledge graph completion and question answering," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, S. Muresan, P. Nakov, and A. Villavicencio, Eds. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 2814–2828. [Online]. Available: https://aclanthology.org/2022.acl-long.201/

[20] A. Miller, A. Fisch, J. Dodge, A.-H. Karimi, A. Bordes, and J. Weston, "Key-value memory networks for directly reading documents," 2016. [Online]. Available: https://arxiv.org/abs/1606.03126

[21] X. Zhang, A. Bosselut, M. Yasunaga, H. Ren, P. Liang, C. D. Manning, and J. Leskovec, "Greaselm: Graph reasoning enhanced language models for question answering," 2022. [Online]. Available: https://arxiv.org/abs/2201.08860

[22] J. Shi, S. Cao, L. Hou, J. Li, and H. Zhang, "TransferNet: An effective and transparent framework for multi-hop question answering over relation graph," in *Proceedings of the 2021 Conference on Empirical*

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:02:57 UTC from IEEE Xplore. Restrictions apply.

Methods in Natural Language Processing, M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, Eds. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 4149–4158. [Online]. Available: https://aclanthology.org/2021.emnlp-main.341/

[23] Y. Sun, L. Zhang, G. Cheng, and Y. Qu, "Sparqa: Skeleton-based semantic parsing for complex questions over knowledge bases," 2020. [Online]. Available: https://arxiv.org/abs/2003.13956

[24] Y. Lan and J. Jiang, "Query graph generation for answering multi-hop complex questions from knowledge bases," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault, Eds. Online: Association for Computational Linguistics, Jul. 2020, pp. 969–974. [Online]. Available: https://aclanthology.org/2020.acl-main.91

[25] R. Das, M. Zaheer, D. Thai, A. Godbole, E. Perez, J. Y. Lee, L. Tan, L. Polymenakos, and A. McCallum, "Case-based reasoning for natural language queries over knowledge bases," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, Eds. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 9594–9611. [Online]. Available: https://aclanthology.org/2021.emnlp-main.755/

[26] R. Yang, H. Liu, E. Marrese-Taylor, Q. Zeng, Y. Ke, W. Li, L. Cheng, Q. Chen, J. Caverlee, Y. Matsuo, and I. Li, "KG-rank: Enhancing large language models for medical QA with knowledge graphs and ranking techniques," in *Proceedings of the 23rd Workshop on Biomedical Natural Language Processing*, D. Demner-Fushman, S. Ananiadou, M. Miwa, K. Roberts, and J. Tsujii, Eds. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 155–166. [Online]. Available: https://aclanthology.org/2024.bionlp-1.13/

[27] S. Li, Y. Gao, H. Jiang, Q. Yin, Z. Li, X. Yan, C. Zhang, and B. Yin, "Graph reasoning for question answering with triplet retrieval," in *Findings of the Association for Computational Linguistics: ACL 2023*, A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds. Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 3366–3375. [Online]. Available: https://aclanthology.org/2023.findings-acl.208/

[28] J. Baek, A. F. Aji, and A. Saffari, "Knowledge-augmented language model prompting for zero-shot knowledge graph question answering," in *Proceedings of the 1st Workshop on Natural Language Reasoning and Structured Explanations (NLRSE)*, B. Dalvi Mishra, G. Durrett, P. Jansen, D. Neves Ribeiro, and J. Wei, Eds. Toronto, Canada: Association for Computational Linguistics, Jun. 2023, pp. 78–106. [Online]. Available: https://aclanthology.org/2023.nlrse-1.7

[29] X. Tan, X. Wang, Q. Liu, X. Xu, X. Yuan, and W. Zhang, "Paths-over-graph: Knowledge graph empowered large language model reasoning," 2025. [Online]. Available: https://arxiv.org/abs/2410.14211

[30] L. Luo, Z. Zhao, C. Gong, G. Haffari, and S. Pan, "Graph-constrained reasoning: Faithful reasoning on knowledge graphs with large language models," 2024. [Online]. Available: https://arxiv.org/abs/2410.13080

[31] M. Dehghan, M. Alomrani, S. Bagga, D. Alfonso-Hermelo, K. Bibi, A. Ghaddar, Y. Zhang, X. Li, J. Hao, Q. Liu, J. Lin, B. Chen, P. Parthasarathi, M. Biparva, and M. Rezagholizadeh, "EWEK-QA : Enhanced web and efficient knowledge graph retrieval for citation-based question answering systems," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, L.-W. Ku, A. Martins, and V. Srikumar, Eds. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 14 169–14 187. [Online]. Available: https://aclanthology.org/2024.acl-long.764/

[32] A. Talmor, J. Herzig, N. Lourie, and J. Berant, "CommonsenseQA: A question answering challenge targeting commonsense knowledge," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, J. Burstein, C. Doran, and T. Solorio, Eds. Minneapolis, Minnesota: Association for Computational Linguistics, Jun. 2019, pp. 4149–4158. [Online]. Available: https://aclanthology.org/N19-1421/

[33] R. Speer, J. Chin, and C. Havasi, "Conceptnet 5.5: An open multilingual graph of general knowledge," 2018. [Online]. Available: https://arxiv.org/abs/1612.03975

[34] Y. Feng, X. Chen, B. Y. Lin, P. Wang, J. Yan, and X. Ren, "Scalable multi-hop relational reasoning for knowledge-aware question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, B. Webber, T. Cohn, Y. He, and Y. Liu, Eds. Online: Association for Computational Linguistics, Nov. 2020, pp. 1295–1309. [Online]. Available: https://aclanthology.org/2020.emnlp-main.99

[35] B. Fatemi, J. Halcrow, and B. Perozzi, "Talk like a graph: Encoding graphs for large language models," in *The Twelfth International Conference on Learning Representations*, 2024.

[36] D. Khashabi, S. Min, T. Khot, A. Sabharwal, O. Tafjord, P. Clark, and H. Hajishirzi, "Unifiedqa: Crossing format boundaries with a single qa system," 2020. [Online]. Available: https://arxiv.org/abs/2005.00700

[37] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," 2023. [Online]. Available: https://arxiv.org/abs/1910.10683

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 16:02:57 UTC from IEEE Xplore. Restrictions apply.